Research Prototype — Not Clinically Deployed

3-Channel Wearable-IMU Stroke C

Participant-level stroke vs. healthy gait discrimination from tri-sensor acceleration-ma

RESEARCH QUESTION

Can participant-level stroke ↔ healthy discrimination be learned from raw 3-channel IMU magnitude windows using deep and classical models?

3-CHANNEL INPUT

- LB Lower Back · L5/Lumbar
- LF Left Foot · swing/clearance
- RF Right Foot · bilateral stance

Source: README.md · docs/DEEP_LEARNING_DEVELOPMENT_PLAN.md · docs/BASELINE_REQUIREMENT

Executive Summary

0.965	0.915	0.914	0.714
Internal OOF AUROC	Internal Balanced Acc.	External RevalExo AUROC	External Balanced Acc.
Expanded · 314 participants	Expanded OOF · 126H / 188S	Locked · 17 participants	Small cohort — descriptive

✅ Current Strengths

- Participant-disjoint 5-fold CV with fold-specific normalization
- Multi-source pooled training (Felius + Voisard + Sint)
- Paired bootstrap Sint gate passed (AUROC Δ = +0.044, CI [0.000, +0.171])
- Locked frozen external test (RevalExo) — no leakage
- Channel occlusion identifies lower-back as most influential

⚠️ Key Limitations

- RevalExo = 17 participants — underpowered for model selection
- No subject-level age/sex linkage across sources
- No direct gait-speed metadata (865 trials — all missing)
- Non-stroke clinical specificity unresolved
- No validated clinical decision threshold

Source: data/processed/population_robustness_matrix.csv · reports/FULL_EXPANDED_PROTOTYPE_BENCHMARK.md · reports/SINT_SENSITIVITY_TRAINING.md · docs/BASELINE_REQUIREMENTS_AUDIT.md

Clinical Motivation

Why Gait Classification Matters Post-Stroke

- **Prevalence:** Stroke is a leading global cause of disability; gait impairment affects the vast majority of survivors.
- **Clinical Gap:** Wearable IMUs enable continuous monitoring, but automated validated classifiers are lacking.
- **Specificity Challenge:** Distinguishing stroke from healthy controls does NOT prove specificity against PD, MS, or COPD.
- **Asymmetric Gait:** Hemiplegia causes bilateral asymmetry across lower-back and bilateral feet.

Known Confounders & Gaps

AGE DISTRIBUTION

0 stroke cases in 18–39 band; 40–59 and 60+ have small cells (≤27 CVA). Healthy age range is wider.

WALKING SPEED

All 865 primary-manifest trials lack recorded walking speed — speed independence is unresolved.

SEVERITY & WALKING AIDS

Fugl-Meyer, FAC scores, and assistive device metadata are unlinked across public datasets.

Input Representation & Sensor Roles

LB Lower Back (L5)

- **Signal:** Trunk acceleration magnitude
- **Physiological role:** Captures pelvic tilt, trunk stability, and lateral sway characteristic of hemiparetic gait
- **Occlusion rank:** **#1** — Mean |ΔP| = **0.188**
- **Removal effect:** Generally *reduces* stroke probability (mean signed Δ = +0.130)
- ⚠ One fold only — cross-fold attribution pending

LF Left Foot

- **Signal:** Foot acceleration magnitude
- **Physiological role:** Foot clearance, swing timing, and push-off force on the paretic or non-paretic side
- **Occlusion rank:** **#3** — Mean |ΔP| = **0.090**
- **Removal effect:** Small mixed effect (mean signed Δ = +0.028)
- ⚠ One fold only — cross-fold attribution pending

RF Right Foot

- **Signal:** Foot acceleration magnitude
- **Physiological role:** Bilateral stance asymmetry; complements left-foot signal for step-timing and loading differences
- **Occlusion rank:** **#2** — Mean |ΔP| = **0.104**
- **Removal effect:** Tends to *increase* stroke probability (mean signed Δ = −0.039)
- ⚠ One fold only — cross-fold attribution pending

Representation contract: tri-axial accelerometer → per-sensor L2 magnitude → 3-channel tensor [LB, LF, RF] → 5-second windows @ 100 Hz → participant-level aggregated prediction. No fabricated channels for incompatible sources.

Source: reports/INCEPTION_OCCLUSION_ANALYSIS.md · data/processed/inception_channel_occlusion_windows.csv · docs/DEEP_LEARNING_DEVELOPMENT_PLAN.md

Data Sources & Dataset Roles

Dataset	Role	N (H / S)	Age Range	Sex	IMU Compat.	Status / Decision
Felius 2024	Core Training	163 (34H / 129S)	Not linked	Incomplete	✅ LB/LF/RF	Active development
Voisard 2025	Core Training	121 (72H / 49S)	18–90 (wider release)	Partial	✅ LB/LF/RF	Active development
Sint Maartenskliniek	Sensitivity Training	30 (20H / 10S)	Not linked	Not linked	✅ Xsens audited	Gate passed — included
RevalExo	External Eval (locked)	17 (7H / 10S)	Cohort means only	Not linked	✅ Adapter validated	Frozen — no fitting

Preprocessing Pipeline

01. Raw IMU
100Hz Acc/Gyro [LB, LF, RF]

02. Filter
Validation & 4th-Order Butterworth

03. Magnitude
L2 Vector Norm Per Sensor

04. Windows
5s (500 Samples) Fixed Segments

05. Group Split
Participant-Disjoint 5-Fold CV

LEAKAGE PREVENTION

Participant IDs never cross fold boundaries. Normalization stats fitted on training windows only. Window overlap never treated as independent subjects.

AGGREGATION UNIT

Window-level probabilities averaged per participant for a single score. All reported AUROC and balanced accuracy computed at the participant level.

Source: notebooks/03_signal_preprocessing_and_windows.ipynb · docs/DEEP_LEARNING_DEVELOPMENT_PLAN.md § 3–4 · scripts/train_sint_sensitivity_inception.py

Validation Design & Leakage Controls

Participant-Disjoint 5-Fold CV

- StratifiedGroupKFold — group = participant ID
- All trials, visits, and windows from a participant stay in the same fold
- Expanded training: 63, 63, 63, 63, 62 held-out participants per fold
- Fold-specific normalization (mean/SD fitted on training windows only)

Source / Class-Balanced Sampling

- WeightedRandomSampler equalises source × label cells
- Prevents dominant source (Felius 163 pts) from overwhelming minority (Sint 30 pts)
- Applied only to training batches — not to validation

Frozen External Lock — RevalExo

- 17 participants excluded from ALL fitting, tuning, and architecture decisions
- Evaluated once per locked model — never used for thresholding or calibration
- Separate validated adapter maps RevalExo signals → LB/LF/RF magnitude tensor
- Sint external examination also kept blind before sensitivity training

Anti-Leakage Checklist

- ❌ Windows from the same person in train AND val
- ❌ Normalization computed on the full dataset
- ❌ Threshold / calibration fitted on RevalExo
- ❌ Overlapping windows treated as independent participants
- ✅ All controls confirmed by audit scripts

Model Comparison

Internal 5-Fold (Felius + Voisard)

Model	AUROC	Bal. Acc.	F1
MiniROCKET + Ridge	**0.972**	**0.936**	**0.956**
Inception CNN	0.962	0.873	0.879
Compact CNN	0.973	0.857	0.845

⚠️ MiniROCKET internal advantage is consistent, but the external comparison (17 participants) cannot confirm superiority.

Frozen RevalExo External (n=17)

Model	AUROC	Brier	95% CI (AUROC Δ vs. Inception)
MiniROCKET	0.886	0.214	−0.129 to +0.167
Inception CNN	0.871	**0.185**	—
Expanded Inception	**0.914**	0.161	—

🚫 Do NOT declare a winner. Paired bootstrap CI −0.129 to +0.167 — interval straddles zero. Keep both as co-primary candidates.

Calibration: Inception internal pooled Brier = 0.120 (ECE-10 = 0.182). MiniROCKET calibrated internal Brier = 0.060 (separate calibration run — not directly comparable to external Brier).

Expanded Pooled Training — Sint Maartenskliniek

0.966
Mean Internal AUROC
5-fold · Felius+Voisard+Sint

0.904
Mean Balanced Accuracy
Internal expanded OOF

0.914
RevalExo AUROC (expanded)
vs. 0.871 original Inception

Why Sint Was Evaluated

- 30 participants (20 healthy / 10 stroke) with audited Xsens → LB/LF/RF mapping
- First used as a blind external examination (AUROC 0.915, Brier 0.097)
- Then tested in a sensitivity experiment before unlocking RevalExo gate

Paired RevalExo Bootstrap Gate

Metric	Delta (Expanded – Original)	95% CI
AUROC Δ	**+0.044**	[0.000, +0.171]
Brier Δ	**−0.022**	[−0.040, −0.002]

✅ **Decision Gate Passed:** Brier CI excludes zero. Sint retained as training candidate. No clinical superiority claim (n=17 only).

External Robustness — Locked RevalExo Evaluation

17
Total Participants
7 Healthy · 10 Stroke

0.914
Expanded AUROC
Participant-level

0.161
Brier Score
Descriptive — not clinical

4/7
Healthy False Positives
at descriptive 0.5 cutoff

Confusion Matrix — Expanded Inception (0.5 cutoff)

	Pred: Healthy	Pred: Stroke
True: Healthy	TN: 3	FP: 4
True: Stroke	FN: 0	TP: 10

All 10 stroke participants correctly classified. 4/7 healthy = false positive.

FALSE POSITIVE PATTERN

4 healthy subjects predicted as stroke (probabilities: 0.52–0.76). Likely older/slower walkers without individual-age confirmation.

CONFIDENCE LIMITATION

n=17 is insufficient for calibrated thresholding, subgroup analysis, or clinical validation. Treat as site/device robustness stress-test only.

Source: data/processed/full_expanded_prototype_revalexo_metrics.csv · data/processed/revalexo_external_error_analysis.csv

Population Robustness Matrix

Scope / Population	N	Healthy	Stroke	AUROC	Brier	Bal. Acc.	Evidence Type
Expanded Internal OOF	314	126	188	**0.965**	0.069	**0.915**	OOF — internal
↳ Felius 2024	163	34	129	0.925	0.083	0.884	OOF — source slice
↳ Sint Maartenskliniek	30	20	10	0.945	0.050	0.950	OOF — source slice
↳ Voisard 2025	121	72	49	0.983	0.054	0.914	OOF — source slice
RevalExo External (locked)	17	7	10	**0.914**	0.161	0.714	Independent external

SOURCE DIVERSITY

3 sources, multiple sites and devices. Internal performance consistent across sources.

DEMOGRAPHIC DIVERSITY ≠ SOURCE DIVERSITY

No linked age/sex for most participants. Age-stratified and sex-stratified results are NOT available.

NON-STROKE SPECIFICITY GAP

Current training population = stroke + healthy only. No test against PD, MS, COPD, fracture.

Demographic Subgroup Gate

Age-Band Coverage (Binary Labels Only)

Age Band	CVA (Stroke)	HS (Healthy)	Stroke AUROC	Status
18–39 (Young)	0	43	N/A	❌ Cannot estimate
40–59 (Middle)	27	15	**0.953 (exploratory)**	⚠️ Small / imbalanced
60+ (Older)	22	15	**0.913 (exploratory)**	⚠️ Small / imbalanced

Required Next Data: ≥20–30 participants per class per band. Mandatory next step is verifying subject-level age/sex linkage across sources.

🚫 Critical Gap: Young Stroke

Zero stroke participants aged 18–39. 43 healthy subjects in this band creates severe age confounding — the model has never seen young stroke gait.

⚠️ Fairness Claim Blocked

Age-generalisation & fairness claims are NOT supported. Sex linkage is also incomplete. Current results reflect exploratory confound analyses only.

Mobilise-D Clinical Specificity Stress-Test

2,315
Participants (visit-level)
COPD · MS · PD · PFF

0.713
Visit-Level Balanced Acc.
22 DMO features · 5-fold CV

0.717
Bout-Level Balanced Acc.
16M bouts · 17 features

Why NOT Pooled Into Training

- CVS release contains **processed digital mobility outcomes (DMO)** from a single back-worn sensor — NOT raw bilateral LB/LF/RF windows
- No stroke or healthy-control cohort exists in CVS
- No representation-matching experiment has been performed
- Pooling would require fabricating missing bilateral channels

Value as Stress-Test Resource

- Clinical DMO features (walking speed, cadence, stride length) distinguish COPD/MS/PD/PFF cohorts with ~0.71 balanced accuracy
- Bout-level vs. visit-level difference is small (+0.004) — no new training pathway justified
- Demonstrates that non-stroke clinical populations produce measurably different mobility signatures
- Required for *future* false-positive specificity testing before clinical deployment

Source: data/processed/mobilise_d_clinical_cohort_benchmark_metrics.json · data/processed/mobilise_d_bout_cohort_benchmark_metrics.json · reports/MOBILISE_D_CLINICAL_COHORT_BENCHMARK.md · reports/MOBILISE_D_CVS_COMPATIBILITY_AUDIT.md

Failure Analysis & Explainability

Per-Dataset Error Breakdown (Internal)

Model	Dataset	Sens.	Spec.	FP	FN
Inception	Felius	0.837	0.853	5	21
Inception	Voisard	0.776	0.958	3	11
MiniROCKET	Felius	**0.984**	0.824	6	**2**
MiniROCKET	Voisard	**0.918**	0.944	4	**4**

0.5 cutoff is descriptive only. Speed metadata missing for error stratification.

Channel Occlusion (Fold-0 Only)

Channel	Mean |ΔP|	Signed Δ	Rank
LB Lower Back	**0.188**	+0.130	🥇 1st
RF Right Foot	0.104	−0.039	🥈 2nd
LF Left Foot	0.090	+0.028	🥉 3rd

⚠️ Single fold only. Cross-fold temporal attribution pending.

RevalExo False Positive Pattern

4/7 healthy classified as stroke (prob: 0.52–0.76). Likely older/slower walkers without individual-age confirmation.

Domain Shift & Speed Confound

CNN sensitivity drops in Voisard vs. Felius; MiniROCKET is more balanced. Missing trial speed prevents speed-stratified validation.

Research Sufficiency vs. Clinical Deployment Gaps

✅ Sufficient for Research Prototype

✓ Binary stroke/healthy discrimination
AUROC 0.965 internal · 0.914 external (RevalExo)

✓ Leakage-controlled validation
GroupKFold · fold-specific normalization · locked external test

✓ Multi-source wearable-IMU benchmark
Felius + Voisard + Sint (n=314) with site diversity

✓ Locked external test
Independent RevalExo site · paired bootstrap gate

✓ Non-stroke clinical stress testing
2,315 Mobilise-D cohort participants

🚫 Gaps Blocking Clinical Deployment

✗ No validated decision threshold
0.5 cutoff is descriptive only — no prospective trial

✗ Subgroup fairness unresolved
No stroke participants aged 18–39; metadata incomplete

✗ Non-stroke specificity untested
Never evaluated against PD, MS, COPD, or fracture gait

✗ Speed independence unconfirmed
865 trials with missing speed — potential confounder

✗ External sample underpowered
n=17 is too small to declare architecture superiority

Source: docs/BASELINE_REQUIREMENTS_AUDIT.md · reports/DATA_SUFFICIENCY_AND_PUBLIC_SOURCE_REVIEW.html · reports/REVISED_MODEL_SELECTION_GATE.md

Final Synthesis

BEST INTERNAL RESULT

0.965

AUROC · 314 participants

Expanded Inception · OOF

BEST EXTERNAL RESULT

0.914

AUROC · 17 participants (locked)

RevalExo · Brier = 0.161

KEY LIMITATION

n=17

External cohort — too small for model selection, clinical validation, or age/sex subgroups

Verified Research Claim

"The current system is a **promising research prototype** for participant-level stroke-versus-healthy gait discrimination across multiple public wearable-IMU sources (Felius + Voisard + Sint; AUROC 0.965 internal). An independent external stress-test on RevalExo (n=17) yields AUROC 0.914. **Age-stratified, sex-stratified, non-stroke clinical specificity, and clinical deployment evidence remain incomplete.** Both Inception and MiniROCKET are co-primary candidates — no winner is declared."

What can be published now: Methodology, internal benchmark, external RevalExo result, limitations, and roadmap.

What must NOT be claimed: Clinical readiness, age fairness, superiority of either model, or speed independence.

Source: data/processed/population_robustness_matrix.csv · reports/FULL_EXPANDED_PROTOTYPE_BENCHMARK.md · reports/REVISED_MODEL_SELECTION_GATE.md ·

Recommended Next Steps

① LOCK BASELINE

Freeze current expanded Inception checkpoint + MiniROCKET as co-primary candidates. Log exact hashes. No further model changes without new external evidence.

② LINK DEMOGRAPHICS

Obtain subject-level age and sex for Felius, Voisard, Sint. Verify linkage against existing participant IDs. Enable age-stratified exploratory analysis.

③ AUDIT MIMU 85/97

Audit the 85-stroke / 97-healthy wearable-MIMU dataset as a potential age-labelled independent cohort. Verify channel compatibility before pooling or evaluation.

④ NON-STROKE SPECIFICITY

Apply frozen model to Mobilise-D-compatible cohort after representation audit. Measure false-positive rate against PD/MS/COPD/fracture — required for any clinical claim.

Additional Priorities (Ordered)

1. Complete cross-fold temporal channel occlusion for Inception
2. Improve external calibration — ECE-10 currently 0.182 (pooled internal)
3. Obtain walking-speed metadata or perform cadence-stratified analysis
4. Pre-specify equivalence margin for model-selection design (e.g., ±0.03 AUROC)
5. Only then: package for clinical prototype evaluation

Data Acquisition Guardrails

- ❌ Do NOT pool Mobilise-D, PiG, or Triaxial without channel-compatibility verification
- ❌ Do NOT assign synthetic stroke labels to healthy-only datasets
- ❌ Do NOT use RevalExo for any fitting, threshold, or calibration decision
- ✅ Any new source must pass blind external audit before sensitivity training

Reproducibility Appendix

Key Artefacts

Type	Path / Filename	Produces
Script	train_sint_sensitivity.py	Sint fold weights
Script	train_full_expanded.py	Full prototype
Script	benchmark_full_expanded.py	RevalExo metrics
Script	build_population_matrix.py	Robustness matrix
Script	benchmark_mobilise_d.py	Mobilise-D DMO
Notebook	08_robust_pooled_training.ipynb	Internal benchmark
Notebook	26_frozen_external_revalexo.ipynb	Locked eval
Notebook	28_revalexo_error_analysis.ipynb	Domain shift

Key Executed Metrics

Metric File	Key Metric	Value
population_robustness_matrix.csv	Expanded OOF AUROC	**0.965**
full_expanded_revalexo_metrics.csv	RevalExo AUROC	**0.914**
sint_revalexo_bootstrap_gate.csv	AUROC Δ 95% CI	[0.00, +0.17]
architecture_comparison.csv	MiniROCKET AUROC	**0.972**
mobilise_d_clinical_cohorts.json	Clinical Bal. Acc.	0.713
age_group_availability.csv	CVA Age 18–39	**0**
INCEPTION_OCCLUSION_ANALYSIS.md	LB Mean |ΔP|	0.188
pooled_inception_calibration.csv	Pooled Brier	0.120

Model weights: `full_expanded_inception_prototype_seed_42.pt` · `minirocket_ridge_fold_*_seed_42.joblib`